CERTIFICATE OF VERIFICATION

Note: This table completed by ~~HERS~~ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

Title 24, Part 6, Section 150.0(o) **Ventilation for Indoor Air Quality.** All dwelling units shall meet the requirements of ANSI/ASHRAE Standard 62.2. Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings, subject to the amendments specified in Section 150.0(o)1. ***Equation and table numbering on this form corresponds to the numbering for that information in the published ANSI/ASHRAE Standard 62.2-2022~~19~~.***

A. Local Mechanical Exhaust - General Information

01	Dwelling Unit Name	
02	Building Type	
03	Total Kitchen Floor Area	
04	Kitchen Average Ceiling Height	
05	Kitchen Total Conditioned Volume	
06	Kitchen Type	
07	Dwelling Unit Total Floor Area	
08	Kitchen Range (Cooking Stove) Fuel Type	

B. Kitchen Exhaust Systems

01	02	03	04	05	06	07	08	09a	09	10a	10	11	12
System Name	Manufacturer Name	System Type	HVI or AHAM Directory Listed Model Number	HVI or AHAM Directory Listed Rated Airflow	HVI or AHAM Directory Listed Sound Rating	Minimum Airflow (defaults to rated airflow)	Operation Schedule	Method of Compliance	Required Minimum Ventilation Rate	Exception to Maximum Sound Rating	Maximum Sound Rating	Compliance Statement for Airflow	Compliance Statement for Sound

C. Continuous Kitchen Exhaust

01	Total Continuous Ventilation Airflow	
02	Required Minimum Continuous Ventilation Airflow	
03	Compliance Statement	

~~C2~~D. Kitchen Range Hood Capture Efficiency Option

01	Manufacturer Name	
02	CEC-Approved Directory Listed Model Number	
03	CEC-Approved Directory Listed Rated Capture Efficiency	
04	Required Minimum Capture Efficiency (*Table 150.0-G*)	
05	Compliance Statement	

~~D~~E. Determination of ~~HERS~~Field Verification Compliance

All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance.

01	

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Documentation Author Company Name:	Date Signed:
Address:	CEA/~~HERS~~ECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Verification is true and correct.
2. I am the certified ~~HERS~~ECC Rater who performed the verification identified and reported on this Certificate of Verification (responsible rater).
3. The installed features, materials, components, manufactured devices, or system performance diagnostic results that require ~~HERS~~field verification identified on this Certificate of Verification comply with the applicable requirements in Reference Appendices RA2, RA3, and the requirements specified on the Certificate of Compliance for the building approved by the enforcement agency.
4. The information reported on applicable sections of the Certificate(s) of Installation (CF2R) signed and submitted by the person(s) responsible for the construction or installation conforms to the requirements specified on the Certificate(s) of Compliance (CF1R) approved by the enforcement agency.
5. I understand that a registered copy of this Certificate of Verification shall be posted, or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
6. I understand that a registered copy of this Certificate of Verification is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to ensure this requirement is accomplished.

Responsible Builder/Installer Name:	Responsible Builder/Installer Signature:	
Company Name: (Installing Subcontractor or General Contractor or Builder/Owner)	Position With Company (Title):	
Address:	CSLB License:	
City/State/Zip:	Phone:	Date Signed:

CF3R-MCH-32-H User Instructions

Section A. Local Mechanical Exhaust - General Information

1. Dwelling Unit Name: This field is filled out automatically and referenced from the MCH-01
2. Building Type: This field is filled out automatically and referenced from the CF1R.
3. Total Kitchen Floor Area: Enter the total floor area for an enclosed kitchen or N/A for a non-enclosed kitchen.
4. Kitchen Average Ceiling Height: Enter the kitchen ceiling height for an enclosed kitchen or N/A for a non-enclosed kitchen.
5. Kitchen Total Conditioned Volume: This field is filled out automatically and calculated based on the kitchen area and ceiling height.
6. Kitchen Type: Enter the type of kitchen (enclosed or non-enclosed).
7. Dwelling Unit Total Floor Area: This field is filled out automatically and referenced from the MCH-01.
8. Kitchen Range Fuel Type: Select the fuel type of the kitchen range: natural gas or electric.

Section B. Kitchen Exhaust System

1. System Name: Enter a unique name for the kitchen exhaust system
2. Manufacturer Name: Enter manufacturer name for the kitchen exhaust system.
3. System Type: Select the type of kitchen exhaust system. Options are vented range hood, downdraft, and other.
4. HVI or AHAM Directory Listed Model Number: Enter the kitchen exhaust system model number matching the installed equipment and HVI or AHAM directory.
5. HVI or AHAM Directory Listed Rated Airflow: Enter the rated airflow listed in the HVI or AHAM directory for the above model number.
6. HVI or AHAM Directory Listed Sound Rating: Enter the sound rating listed in the HVI or AHAM directory for the above model number.
7. Minimum Airflow (defaults to rated airflow): Defaults to rated airflow from HVI directory, but editable if exhaust system minimum airflow rate is less than HVI listed value.
8. Operation Schedule: Select the kitchen exhaust system operation schedule. Options are demand control and continuous.

9a. Method of Compliance: Select the method of compliance. Options are airflow and capture efficiency.

9. Required Minimum Ventilation Rate (if demand controlled): This field is filled out automatically and is calculated based on the system operation schedule and type, and kitchen type and volume, and Table 150.0-E and Table 150.0-G. This field is only used for demand control exhaust systems. Continuous exhaust required minimum ventilation rate is determined in Section D.

10a. Exception to Maximum Sound Rating: User select: No Exception or Remote mounted fan with min. 4-ft of duct between fan and intake grille.

10. Maximum Sound Rating: This field is filled out automatically and is calculated based the system operation schedule and minimum airflow.
11. Compliance Statement for Airflow: This field is filled out automatically based on the installed system listed airflow rate and minimum required ventilation rate. This field only determines compliance using airflow ratings for demand-controlled kitchen exhaust systems. Continuous system ventilation rate compliance is determined in Section D. Vented range hoods utilizing the capture efficiency rating for compliance is determined in Section E.
12. Compliance Statement for Sound. This field is filled out automatically based on the installed system listed sound rating and maximum sound rating allowed.

Section C. Continuous Kitchen Exhaust

1. Total Continuous Ventilation Airflow: This field is filled out automatically and is equal to the sum of the HVI listed airflow for all continuously operated kitchen exhaust systems.
2. Required Minimum Continuous Ventilation Airflow: This field is filled out automatically and is equal to five times the enclosed kitchen volume.
3. Compliance Statement: This field is filled out automatically and is based on the total installed continuous ventilation airflow and the required minimum continuous ventilation airflow.

Section ~~C2~~D. Kitchen Range Hood Capture Efficiency Option

Note: This table is used only when complying with local exhaust requirements by utilizing the capture efficiency rating instead of the airflow rating.

1. Manufacturer Name: Enter manufacturer name for the kitchen range hood.
2. CEC-Approved Directory Listed Model Number: Enter the kitchen range hood model number matching the installed equipment and a CEC-approved directory listing.
3. CEC-Approved Directory Listed Rated Capture Efficiency: Enter the rated capture efficiency in the CEC-approved directory for the above model number.
4. Required Minimum Capture Efficiency: This field is filled out automatically and is determined by the dwelling unit square footage, kitchen range fuel type, and Table 150.0-G.
5. Compliance Statement. This field is filled out automatically based on the installed system listed capture efficiency rating and required minimum capture efficiency.

Section ~~D~~E. Determination of ~~HERS~~ECC Verification Compliance

1. This field is filled out automatically based on all verification protocol requirements in this document showing compliance.

Title 24, Part 6, Section 150.0(o) **Ventilation for Indoor Air Quality.** All dwelling units shall meet the requirements of ANSI/ASHRAE Standard 62.2. Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings subject to the amendments specified by Title 24, Part 6, Section 150.0(o)1. ***Equation and table numbering on this form corresponds to the numbering for that information in the published ANSI/ASHRAE Standard 62.2-2022~~19~~.***

A. Local Mechanical Exhaust - General Information		
01	Dwelling Unit Name	<<Calculated field, referenced data from MCH-01, "Dwelling Unit Name" (A01) or Prescriptive CF1R; else if information is not available then allow user input.>>
02	Building Type	<< reference data from CF1R; else if information is not available then allow user input. Allowed values = single family detached, or single family attached>>
03	Total Kitchen Floor Area	<<User Entered Value (XX.XX), if A06 = Non-Enclosed and [C03 = VentedRangeHood & C08 = Demand Control for ALL systems], then allow N/A>>
04	Kitchen Average Ceiling Height	<<User Entered Value (XX.XX), if A06 = Non-Enclosed and [C03 = VentedRangeHood & C08 = Demand Control for ALL systems], then allow N/A>>
05	Kitchen Total Conditioned Volume	<<calculated value, "Kitchen Floor Area (A03)" * "Kitchen Average Ceiling Height" (A04) (XX.XX); else if A03 or A04 = N/A, then value = N/A>>
06	Kitchen Type	User Entry, selections (Enclosed or Non-Enclosed)
07	Dwelling Unit Total Floor Area	<<Calculated field, referenced data from MCH-01, "Dwelling Unit Total Conditioned Floor Area" (field A03) or Prescriptive CF1R; else if information is not available then allow user input >>
08	Kitchen Range (Cooking Stove) Fuel Type	<<User Entered Value; Selections = (Electric, Natural Gas)>>

B. Kitchen Exhaust Systems													
01	02	03	04	05	06	07	08	09a	09	10a	10	11	12
System Name	Manufacturer Name	System Type	HVI or AHAM Directory Listed Model Number	HVI or AHAM Directory Listed Rated Airflow	HVI or AHAM Directory Listed Sound Rating	Minimum Airflow (defaults to rated airflow)	Operation Schedule	Method of Compliance	Required Minimum Ventilation Rate	Exception to Maximum Sound Rating	Maximum Sound Rating	Compliance Statement for Airflow	Compliance Statement for Sound

<<User Entered Value up to 50 characters>>	<<User Entered Value up to 50 characters>>	<<User Entered Value; Selections = (Vented Range Hood, Downdraft, Other)>>	<<User Entered Value up to 50 characters>>	<<User Entered Value; (XXXX.XX)>>	<<User Entered Value; (XX.XX)>>	<<Defaults to B05 otherwise, User Entered Value; (XXXX.XX); Not to exceed B05 (rated airflow)>>	<<User Entry; Selections = (Demand Control, Continuous)>>	<<If B08=Continuous, then report N/A; Else user select from list: *Airflow; or *Capture Efficiency>>	<<if B09a = Capture Efficiency, then result = "NA – See Table ~~C2~~D."; Elseif B09a = Airflow, then calculate the following as default: If the following three conditions are met: (1) B08=Demand Control, (2) B03=Vent Range Hood, and (3) A08=Electric, AND If A07 > 1500, then result = 110; Elseif A07 = 1,000-1,500, then result = 110; Elseif A07 = 750-1,000, then result = 130;	<<default value = No Exception, but allow user to override and select: *Remote mounted fan with min. 4-ft of duct between fan and intake grille>>	<<If C10a = No Exception, then value = N/A; ElseIf Continuous, then value = "1 sone"; ElseIf Demand Control and B07 ≤ 400 cfm, then value = "3 sone"; Else value = "N/A">>	<<If B09a = Capture Efficiency, then result = N/A; ElseIf B08 = 'Demand Control', and B05 (HVI Directory Listed Rated Airflow) ≥ B09 (Required Minimum Ventilation Rate), then display text: "Complies"; ElseIf B08 = Continuous, then display text: "N/A"; else display text: "Does Not Comply" >>	<<If B06 ≤ B10 or B10 = N/A, then display text: "Complies"; else display text: "Does Not Comply" >>

									ElseifA07 = < 750, then result = 160; End if; Elseif the following three conditions are met: (1) B08=Demand Control, (2) B03=Vent Range Hood, and (3) A08 = Natural Gas, AND If A07 > 1,500, then result = 180; Elseif A07=1,000-1,500, then result = 250; Elseif A07 = 750-1,000, then result = 280; *Elseif A07 = <750, then result = 280;* End if; Elseif the followi				

									ng three conditions are met: (1) B08=Demand Control, (2) A06 = Enclosed, and B03 = Other or Downdraft, then result is the lesser of 300 cfm or (5*(A05/60)); Elseif the following three conditions are met: (1) B08=Demand Control, (2) A06 = Nonenclosed, and (3) B03 = Other or Downdraft, then result = 300; Else result = "N/A – See Table C."				

C. Continuous Kitchen Exhaust		
01	Total Continuous Ventilation Airflow	<<Result = Sum(B05 for all B08 = Continuous) {sum ‘listed rated airflow’ for all continuously operated fans}; Else result = “N/A”>> {want this entry to be N/A if there are no continuously operated fans}
02	Required Minimum Continuous Ventilation Airflow	<<If C01 = N/A, then result = “N/A”, Else result = 5*A05/60>>
03	Compliance Statement	<<If C01 = N/A, then result = “N/A”; Else if C01 ≥ C02 then result = “Complies”, else result = “Does Not Comply”

~~C2~~D. Kitchen Range Hood Capture Efficiency Option << if B09a = Capture Efficiency, then display table; else display section header and standard “this section does not apply” message>>			
01	Manufacturer Name	<<User Entered Value up to 50 characters>>	*
02	CEC-Approved Directory Listed Model Number	<<User Entered Value up to 50 characters>>	
03	CEC-Approved Directory Listed Rated Capture Efficiency (%)	<<User Entered Value; (XX.XX)>>	
04	Required Minimum Capture Efficiency (%)	<<Result = “50” for A08 = Electric and A07 > 1500; “50” for A08 = Electric and A07 is 1000 – 1500; “55” for A08 = Electric and A07 is 750 – 1000; “65” for A08 = Electric and A07 < 750; “70” for A08 = Natural Gas and A07 > 1500; “80” for A08 = Natural Gas and A07 is 1000 – 1500; “85” for A08 = Natural Gas and A07 is 750 – 1000; “85” for A08 = Natural Gas and A07 < 750>>	
05	Compliance Statement	<<If ~~C2~~D-03 ≥ ~~C2~~D-04, then display text: "Complies”; else display text: "Does Not Comply" >>	

~~D~~E. Determination of ~~HERS~~ECC Verification Compliance All applicable sections of this document shall indicate compliance with the specified verification protocol requirements in order for this Certificate of Verification as a whole to be determined to be in compliance.	
01	<<if (1)B12 = complies for all rows (exhaust systems), and (2) if B11, C03 or ~~C2~~D-05 = complies, then Result = “Complies: All specified verification protocol requirements on this document are met”; else Result = “Does not comply: One or more specified verification protocol requirements on this document are not met”>>